

8ai	initial speed is zero, constant acceleration, straight line motion (any 2)
8aii1	$s = \frac{1}{2}at^2$ $0.79 = \frac{1}{2}(9.81)t^2$ $t = 0.401 \text{ s (accept 2 or 3sf)}$
8aii2	Distance travelled by end of time interval = 90 cm $0.90 = \frac{1}{2}(9.81)t^2$ $t = 0.428 \text{ s}$ time interval = $0.428 - 0.401 = 0.027 \text{ s}$
8aii3	$\% \text{ difference} = \frac{0.033 - 0.027}{0.033} = 18\%$ Since the percentage difference is larger than 5%, it is significant, hence the test does not confirm this time. (accept different stated criteria).
8aiii	The <u>speed/acceleration</u> of a less dense ball affected by air resistance will be <u>lesser</u> than the denser one as net force acting on the less dense ball decreases. Hence, <u>length of image is shorter</u> .
8bi	force = rate of change of momentum (allow symbols if defined)
8bii1	change in momentum of the ball, $\Delta p = 140 \times 10^{-3} \times (5.5 + 4.0)$ $= 1.33 \text{ kg m s}^{-1}$
8bii2	$F_{\text{net}} = 1.33 / 0.04$ $= 33.25 \text{ N}$ $= 33 \text{ N}$ Average contact force on bar due to ball $= F_{\text{net}} + W_{\text{ball}}$ $= (1.33/0.04) + (9.81 \times 140 \times 10^{-3})$ $= 34.6$ $= 35 \text{ N}$
8bii3.	Taking moments about B $(35 \times 0.75) + (0.450 \times 9.81 \times 0.25) = F_A \times 0.20$ $F_A = 137 \text{ N}$
8biii	The change in momentum of the ball on hitting the bar decreases, the duration of contact increases,

hence, average contact force on bar at point D decreases.

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Half life is the average time taken for the activity